

# TASK –2

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No:** 1

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**Branch:** CSM

**Section:** B

**Date:**

## 1. TITLE

JavaScript-Based Student Profile Application

## 2. AIM

To develop a Student Profile Application that accepts student details and dynamically displays them using JavaScript classes, DOM manipulation, dynamic HTML elements, and event handling.

## 3. OBJECTIVE

- To understand JavaScript classes and objects.
- To implement DOM manipulation for dynamic content creation.
- To use `createElement()` and `appendChild()` for generating HTML elements.
- To apply event handling using `addEventListener()`.
- To build a simple interactive Student Profile Application.

## 4. THEORY

- JavaScript Classes allow defining reusable templates for objects.
- DOM Manipulation enables dynamic creation and modification of HTML elements.
- Event Handling allows responding to user actions like button clicks.
- Methods like `createElement()`, `textContent`, and `appendChild()` are used to generate and insert elements dynamically.

Flow of application: User Input → JavaScript Class → Object Creation → DOM Manipulation → Dynamic Profile Display

## 5. ALGORITHM / PROCEDURE

1. Create an HTML file with input fields for Name, Roll Number, Department, and CGPA.
2. Add a button to trigger profile display.
3. Define a `Student` class in JavaScript with properties.
4. Capture input values using DOM methods.
5. Create a `Student` object using the class constructor.
6. Dynamically generate HTML elements (`div`, `p`, `h2`) using `createElement()`.
7. Insert values into elements using `textContent`.
8. Append elements to the webpage using `appendChild()`.
9. Implement event handling with `addEventListener()` for button click.
10. Test with multiple student inputs.

## 6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Student Profile</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
    text-align: center;
    padding: 30px;
  }

  .container {
    width: 400px;
    margin: auto;
    background-color: white;
    padding: 25px;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
  }

  input {
    width: 90%;
    padding: 10px;
    margin: 8px;
    border: 1px solid #ccc;
    border-radius: 5px;
  }

  button {
    padding: 10px 20px;
    background-color: #007bff;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    margin-top: 10px;
  }

  button:hover {
    background-color: #0056b3;
  }

  .profile {
    margin-top: 25px;
    padding: 20px;
  }
```



```
// Creating Student class
class Student {

    constructor(name, rollNumber, department, cgpa) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.department = department;
        this.cgpa = cgpa;
    }

}

// Selecting the button
const button = document.getElementById("displayBtn");

// Adding click event
button.addEventListener("click", function() {

    // Getting values from input fields
    const name = document.getElementById("name").value;
    const rollNumber = document.getElementById("roll").value;
    const department = document.getElementById("department").value;
    const cgpa = document.getElementById("cgpa").value;

    // Creating Student object
    const student = new Student(
        name,
        rollNumber,
        department,
        cgpa
    );

    // Selecting profile container
    const profileContainer =
        document.getElementById("profileContainer");

    // Clear previous profile
    profileContainer.innerHTML = "";

    // Creating profile div dynamically
    const profile = document.createElement("div");
```

```
profile.className = "profile";

// Creating heading
const heading = document.createElement("h2");

heading.textContent = "Student Profile";

// Creating paragraphs
const namePara = document.createElement("p");

namePara.textContent = "Name      : " + student.name;

const rollPara = document.createElement("p");

rollPara.textContent = "Roll No    : " + student.rollNumber;

const departmentPara = document.createElement("p");

departmentPara.textContent =
    "Department : " + student.department;

const cgpaPara = document.createElement("p");

cgpaPara.textContent = "CGPA      : " + student.cgpa;

// Adding elements to profile
profile.appendChild(heading);

profile.appendChild(namePara);

profile.appendChild(rollPara);

profile.appendChild(departmentPara);

profile.appendChild(cgpaPara);

// Adding profile to webpage
profileContainer.appendChild(profile);

});
```

```

    </script>
</body>
</html>

```

## 7. CODE EXPLANATION

| Code                      | Explanation                                        |
|---------------------------|----------------------------------------------------|
| class Student {}          | Defines a Student class with properties.           |
| document.getElementById() | Accesses input fields and button.                  |
| addEventListener("click") | Handles button click event.                        |
| createElement()           | Dynamically creates HTML elements.                 |
| textContent               | Inserts text into elements.                        |
| appendChild()             | Adds elements to the DOM.                          |
| innerHTML=""              | Clears previous profile before displaying new one. |

## 8. TEST CASES AND EXECUTION DATA

| Test Case | Input                                                              | Expected Result                   | Actual Result      | Status |
|-----------|--------------------------------------------------------------------|-----------------------------------|--------------------|--------|
| TC-01     | Name: Mohini,<br>Roll:<br>A24126552110,<br>Dept: CSM,<br>CGPA: 9.0 | Profile<br>displayed<br>correctly | Works<br>correctly | pass   |

|       |                                      |                                |                 |      |
|-------|--------------------------------------|--------------------------------|-----------------|------|
| TC-02 | Empty Name field                     | Profile should not display     | Works correctly | Pass |
| TC-03 | Enter multiple profiles sequentially | Each profile replaces previous | Works correctly | Pass |
| TC-04 | Enter CGPA as decimal                | CGPA displayed correctly       | Works correctly | Pass |
| TC-05 | Button hover                         | Button color changes           | Works correctly | Pass |

## 9.OUTPUT

File C:/Users/reddy/OneDrive/Desktop/fs%20lab/TASK21.HTML

The screenshot displays a web application interface for a 'Student Profile'. The title 'Student Profile' is centered at the top in a large, bold, black font. Below the title, there are four input fields, each with a light gray border and rounded corners. The first field is labeled 'Enter Name', the second 'Enter Roll Number', the third 'Enter Department', and the fourth 'Enter CGPA'. All labels are in a light gray font. At the bottom of the form, there is a blue button with white text that reads 'Display Student Profile'. The entire form is set against a light gray background.



## Student Profile

Display Student Profile

### Student Profile

Name : REDDY MOHINI  
Roll No : a24126552110  
Department : CSM  
CGPA : 9.0

## 10.RESULT

Thus, a **JavaScript-based Student Profile Application** was successfully developed. The application demonstrates object-oriented programming, DOM manipulation, dynamic HTML generation, and event handling.